Can a Fisher/Cramér-Rao Bound Predict θ's Empirical Recovery Scatter?

Full-covariance CRB vs. real pixel-likelihood fits, aberrated wavefront

Contents

1 Motivation

Looking at 2D residual plots of $\hat{\theta}_{\text{best}} - \theta_{\text{true}}$ for the `confocal_random_zernike` (aberrated-wavefront) dataset, the scatter looks Gaussian. That raises the obvious question: can a Fisher-information / Cramér-Rao bound predict that spread analytically, rather than only measuring it empirically?

This was tried once before and got stuck. From `notes/joint_profile_inference.tex` (a sibling investigation, §2026-07-20): a Laplace approximation using the *diagonal* of a local finite-difference Hessian at θ's MAP was found **66–220**$\times$ **tighter** than the true empirical scatter across 50 shots. That work diagnosed why, and flagged two things that were never actually done:

1. θ's failure mode is a *ridge* – a continuous, weakly-identified direction (near-zero Fisher eigenvalue), not discrete multimodality like ϕ's. A ridge is compatible with local Fisher/CRB validity *if* it is close to linear/quadratic near the truth – but whether it actually is was flagged as unchecked (Future Work item 0a in that file) and never verified.
2. The failed attempt used a **diagonal-only** Hessian, discarding exactly the off-diagonal correlations a ridge produces. `crb_signal.py` already has `build_joint_fisher()` (full 17×17 joint Fisher over $(\phi, \theta_{z0}, \theta_{z100})$, pixel-likelihood mode, all cross terms) and `full_per_shot_covariance()` (inverts the *full* matrix) – but the latter was dead code, never called from that module's own driver. Nobody had tried the full-covariance version.

This note finishes that specific, previously-flagged, previously-skipped check, reusing the existing (but previously unwired) machinery rather than re-deriving anything. Scope, by explicit direction: **one dataset** (the aberrated `confocal_random_zernike` case), matching CRB to empirical data. A Gaussian-beam-vs-aberrated comparison, and the downstream connection to the β (signal) CRB, are deliberately deferred.

2 Setup

Dataset: `data/R20_N200_A1000000_..._psmapconfocal_random_zernike` (20 runs $\times$ 200 shots, 10^6 atoms/shot, aberrated wavefront via `phase_space_grids.py`'s interpolated-PSMAP path). Cloud prior: independent Gaussian on each of the 8 kinematic components (position, velocity, position spread, velocity spread, in x and y), $\sigma = 10\,\mu$m (or the matching velocity unit) for all 8. Evaluator geometry: **exactly** what `non_phase_shear/analysis/generate_kinematic_estimates.py`'s `best` (pixel-likelihood) fit uses – `bins=32`, tight half-range 1.54×10^{-3} m, `pixel_n_gh=8` – so the CRB is evaluated on the identical pixel/bin configuration that produced the empirical numbers it is compared against. Using a different geometry would make the comparison apples-to-oranges before it starts.

All CRB quantities below are evaluated at the *true* $(\phi, \theta_{z0}, \theta_{z100})$ for each shot – the standard Cramér-Rao convention, and also convenient here since it needs no extra optimizer run (the empirical $\hat{\theta}_{\text{best}}$ values already come from a separate, already-run fit).

Implementation: `python-scripts/theta_crb_vs_empirical.py` (new), importing `crb_signal.py`'s Fisher-matrix code rather than duplicating it.

3 Step 1: Is the ridge direction actually linear/quadratic?

For 10 real shots (`run_000`), built the full 17×17 joint Fisher matrix H at the true $(\phi, \theta_{z0}, \theta_{z100})$, found its smallest eigenvalue $\lambda_{\min}$ and eigenvector v (the ridge direction; $\sigma_{\text{ridge}} = 1/\sqrt{\lambda_{\min}}$), and profiled the *real* Poisson NLL along v at $k\sigma_{\text{ridge}}$ for $k \in \{0.5, 1, 2, 4, 8\}$, comparing to the Fisher-implied quadratic $\frac{1}{2}\lambda_{\min}(k\sigma_{\text{ridge}})^2$.

Result: not cleanly quadratic. The ratio (actual ΔNLL / quadratic ΔNLL) at $k = 0.5$–1 (the scale that actually matters – real fluctuations live near 1σ, not 8σ) ranged from -2.9 **to** $+4.3$ across the 10 shots – including several shots where the sign itself flips (the true NLL is briefly *lower*, not higher, than at the anchor point, in the direction the Fisher matrix calls "increasing"). The ratio does converge toward 1 at larger k (e.g. 0.80–1.23 at $k = 8$), consistent with the quadratic approximation eventually dominating far from the anchor, but that is not the regime real shot noise samples.

This is a genuine, unresolved finding, not swept under the rug: the single worst-conditioned direction is *not* well-approximated by the local quadratic at the scale that matters. And yet – see §4 – the *aggregate*, full 8-dimensional marginal covariance built from the same local Fisher matrix matches the empirical scatter well. The most likely reconciliation: v (found by minimizing over the full 17-dimensional $(\phi, \theta_{z0}, \theta_{z100})$ space) is a mixed direction, not aligned with any single physical θ component's own marginal-variance direction, so a 2–4× local mis-estimate of curvature along v does not necessarily translate into a comparable error in any individual component's marginal variance after the Schur-complement/inversion mixes all 17 directions together. This is a hypothesis, not something checked here – flagged in §7 as follow-up.

4 Step 2+3: Full-covariance CRB vs. empirical RMSE

For the same 10 shots, computed `full_per_shot_covariance(H)` and extracted $\sigma_{\theta_{z0}}$, $\sigma_{\theta_{z100}}$ (marginal std per component), aggregated as root-mean-square across shots. Compared against the empirical RMSE of $\hat{\theta}_{\text{best}} - \theta_{\text{true}}$, pooled over z0 and z100, from the 13 completed runs of the real `confocal_random_zernike` job available at the time of this check (5200 shot/z-slice points).

One data-quality issue found and handled: the raw empirical RMSE for σ_{vy} was 2.33×10^{-6}, a 3.46× mismatch against the CRB's 6.75×10^{-7} – while every other component matched to a few percent. Investigated directly rather than accepted: of 5200 points, a single shot has a σ_{vy} residual of 1.6×10^{-4} (161 μm) – roughly 70× larger than the next-largest residual in that component and wildly inconsistent with the rest of the distribution (median |residual| = 4.4×10^{-7}, matching σ_{vx}'s median almost exactly). RMSE is quadratic in residuals, so this one likely optimizer failure (an L-BFGS-B run landing in a bad basin for that one shot – not a CRB or model problem) was enough to dominate the whole-sample RMSE by itself. Excluding it (a MAD-based robust-outlier filter, $> 20\sigma_{\text{robust}}$ in any component; 1/5200 points excluded) brings σ_{vy}'s empirical RMSE to 6.77×10^{-7}.

Component	CRB (avg $z0$,$z100$)	Empirical RMSE (outlier excl.)	Ratio
μ_{x0}	9.655×10^{-6}	9.816×10^{-6}	1.017
μ_{y0}	9.655×10^{-6}	9.531×10^{-6}	0.987
μ_{vx0}	2.471×10^{-6}	2.511×10^{-6}	1.016
μ_{vy0}	2.473×10^{-6}	2.437×10^{-6}	0.985
σ_x	9.500×10^{-6}	9.912×10^{-6}	1.043
σ_y	9.461×10^{-6}	9.892×10^{-6}	1.046
σ_{vx}	6.767×10^{-7}	6.861×10^{-7}	1.014
σ_{vy}	6.745×10^{-7}	6.771×10^{-7}	1.004

Table 1: Full-covariance theta CRB vs. empirical RMSE (outlier-excluded), all 8 kinematic components, confocal_random_zernike dataset. All values in metres (or the matching velocity unit). Every component agrees to within ±5%.

Result: the full-covariance CRB matches the empirical scatter to within a few percent, across all 8 components, once one known bad fit is excluded. This directly refutes the presumption (reasonable, given the prior 66–220× failure) that Fisher/CRB simply cannot predict θ's recoverability – it was never actually tried with the full covariance before now.

5 Extension: the unaberrated (Gaussian-beam) baselines

The above was scoped to the aberrated dataset only; this section repeats §4's CRB-vs-empirical comparison (Step 2+3 only, not the ridge check) for the two already-fit, unaberrated (`CONFOCAL_FINE`) baseline datasets already saved in `non_phase_shear/results/` (`kinematic_estimates_{1e6,1e8}_N10_shots50` – no new data generation, reusing `python-scripts/theta_crb_gaussian_baseline.py`.

Data-availability caveat for `1e6`. Its source dataset is no longer on disk (the only remaining 1e6-atom, 50-shots/run directory has a different cloud-width prior than what the saved `true_theta` values actually show, so it isn't the same source). The saved JSON has `true_theta` and photon-count totals (n_{tot}) – both of which `build_joint_fisher` needs – but not the true ϕ. Since ϕ is drawn independently of θ (`phi0_mode=random`), this uses a random $\phi \sim U(0, 2\pi)$ substitute per shot instead. `1e8` has its exact source data (`data/R40_N50_A100000000_..._f0.3000`, confirmed by matching `true_theta`/photon-count against its own file attrs), so its numbers use the real per-shot ϕ, no substitution.

Component	CRB (1e6)	Empirical (1e6)	Ratio
μ_{x0}	9.644×10^{-6}	1.017×10^{-5}	1.054
μ_{y0}	9.644×10^{-6}	9.676×10^{-6}	1.003
μ_{vx0}	2.500×10^{-6}	2.638×10^{-6}	1.055
μ_{vy0}	2.501×10^{-6}	2.499×10^{-6}	0.999
σ_x	9.389×10^{-6}	1.019×10^{-5}	1.086
σ_y	9.367×10^{-6}	9.990×10^{-6}	1.067
σ_{vx}	6.676×10^{-7}	7.053×10^{-7}	1.056
σ_{vy}	6.699×10^{-7}	6.909×10^{-7}	1.031

Table 2: `1e6` baseline (unaberrated, ϕ substituted – see caveat above): matches to within ±9%, similar quality to the aberrated result.

Component	CRB (1e8)	Empirical (1e8)	Ratio
μ_{x0}	7.616×10^{-6}	9.279×10^{-6}	1.218
μ_{y0}	7.656×10^{-6}	9.295×10^{-6}	1.214
μ_{vx0}	1.972×10^{-6}	2.402×10^{-6}	1.218
μ_{vy0}	1.982×10^{-6}	2.408×10^{-6}	1.215
σ_x	6.262×10^{-6}	9.877×10^{-6}	**1.577**
σ_y	6.304×10^{-6}	9.822×10^{-6}	**1.558**
σ_{vx}	4.297×10^{-7}	6.703×10^{-7}	**1.560**
σ_{vy}	4.205×10^{-7}	6.731×10^{-7}	**1.601**

Table 3: `1e8` baseline (unaberrated, real ϕ, no substitution): a real, non-trivial gap – means off by ∼21%, widths off by ∼56–60%. Confirmed this is not a numerical artifact: the CRB is stable under h_θ from 10^{-6} to 10^{-8} (only degrading at 10^{-9}, floating-point roundoff), and the empirical residual distribution is broadly heavy-tailed (max $\approx 4\times$ median, consistent across all 8 components) rather than dominated by one outlier shot as in §4.

Interpretation. `1e6` (both aberrated and unaberrated) and the aberrated case both match the CRB to a few percent. `1e8` does not – empirical scatter is systematically 1.2–1.6× the CRB, worst for the width parameters. This is not swept under the rug, and a plausible explanation already exists in prior work rather than needing to be invented here: `notes/joint_profile_inference.tex` (§2026-07-20, a different optimizer but the same underlying pixel likelihood) documented that θ's optimisation *"genuinely struggles at* 10^8*"* – even warm-started at the exact true θ, a conditional solve "wandered/ plateaued... never approaching tolerance" – attributed to the likelihood surface becoming legitimately much sharper at high photon count ($|\nabla_\theta \mathrm{NLL}|$ exceeding the NLL itself). If `fit_theta_best`'s L-BFGS-B is similarly not fully converging at 10^8 (not checked here – see §7), that would inflate the *empirical* RMSE above the true CRB floor without the CRB itself being wrong – consistent with the direction of the observed gap (empirical > CRB, not the reverse) and with 1e6 (where the same prior work reported convergence was fine) matching cleanly.

6 Framing: this is a single-shot comparison, and that's intentional

θ is a fresh, independent nuisance parameter drawn from the prior *every shot* – it is not a shared, estimated-once parameter like $\beta = (A_s, A_c)$. Consequently, pooling more shots does *not* reduce the true per-shot θ RMSE (there is nothing to pool: shot i's θ_i tells you nothing about shot j's θ_j). What more shots *do* improve is the *measurement precision of the empirical RMSE estimate itself* – with 5200 points the empirical RMSE's own relative uncertainty is $\sim 1/\sqrt{2 \times 5200} \approx 1\%$, tight enough to distinguish the observed few-percent CRB agreement from noise. This is why this comparison uses many shots (for a trustworthy *measurement*) while still being a fundamentally single-shot statement (the CRB itself is a per-shot quantity, not one that shrinks with N_{shots}) – unlike β's CRB, which genuinely does improve with shot count (`crb_signal.py`'s `crb_from_j/simple_scaling_crb`).

7 Open questions / future work

- **Reconcile §3 and §4.** Check what the worst-eigenvalue direction v actually projects onto in θ-component space, for a few shots – confirm (or refute) the "mixed direction, doesn't dominate any single marginal" hypothesis directly rather than leaving it as speculation.
- **Diagnose the 10^8 gap** (§5): check whether `fit_theta_best`'s L-BFGS-B is actually converging (`res.nit/res.success`, and/or a tighter `maxiter` / warm-start-from-truth spot check) at 10^8 atoms specifically. If convergence is the cause, this is an estimator-quality issue, not a CRB-validity issue – worth confirming directly rather than resting on the plausible-but-unverified analogy to the joint-solve's documented 10^8 struggles.
- **Exact (not ϕ-substituted) `1e6` baseline** and a dedicated apples-to-apples aberrated-vs-Gaussian comparison at matched N: the `1e6/1e8` baselines used here are convenient (already fit, no new compute) but neither is parameter-matched to `confocal_random_zernike` as closely as `data/R80_N200_A1000000_..._f0.3000` would be (exact n_{atoms}/cloud-prior/shot-count match, already on disk, just never run through `generate_kinematic_estimates.py`) – worth doing if a tight, controlled aberration-only comparison is needed later.
- **Connect to the β (signal) CRB.** Propagate whatever aberrated-vs-Gaussian theta-CRB difference is established through `crb_signal.py`'s existing $j_{\mathrm{eff}} \to$ beta-CRB machinery to get an actual quantitative bound on how much wavefront aberration/curvature costs in signal-extraction precision – the original motivating question for this whole line of work.

- The current check used only `run_000`'s first 10 shots for the CRB side (cheap: ~ 1 minute for all of §3–4 combined). Widening this to more shots/runs would tighten the CRB-side estimate too (currently it isn't the limiting source of noise in the comparison, but worth checking as the empirical run finishes all 20 runs).